Classifying Stimulus Familiarity from Neural Spike Trains

Neuroinformatics - Prof. Gordon Pipa - WiSe 2025

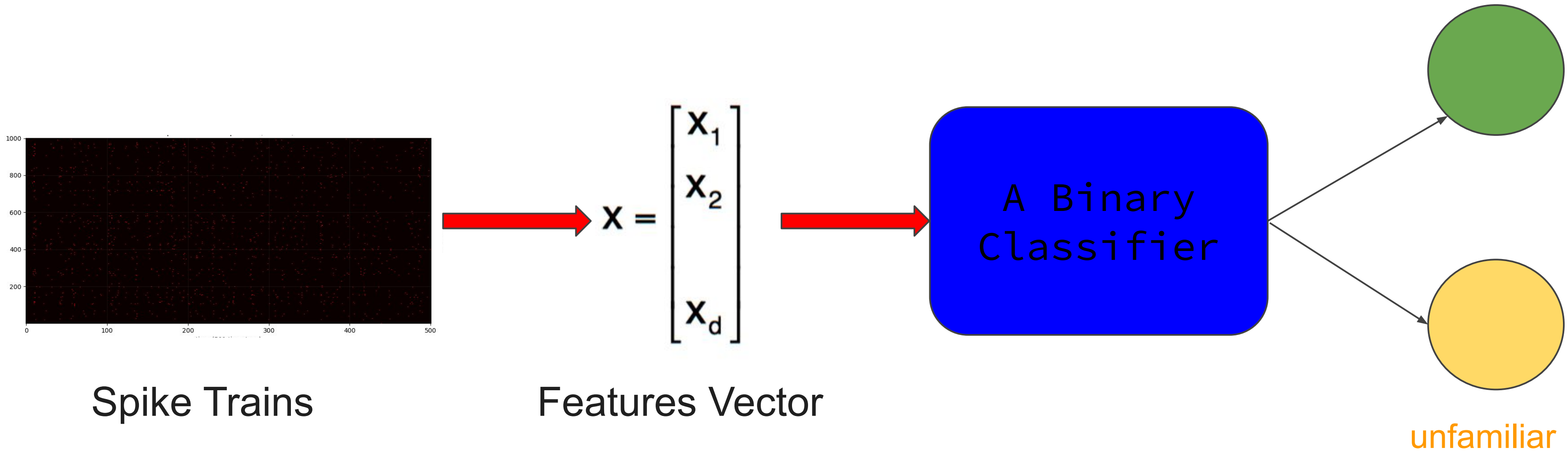
Author: Esma Hodzic

Matrikel-Nr.: 996692

Task Overview

Binary classification: familiar vs unfamiliar stimulus

Data: spike trains from multiple neurons (shape: 800 samples × 1000 neurons × 501 timesteps)
Goal: extract meaningful neural features → train classifiers → evaluate and compare



Feature Extraction (Neural Coding Methods)

Rate coding:

Start

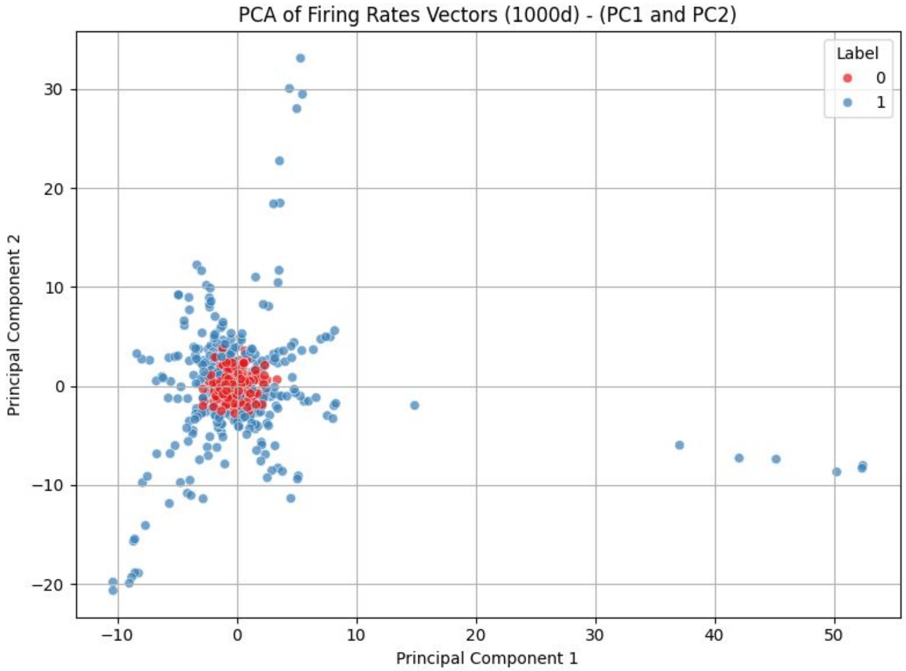
spike train per neuron → mean firing rate (fr) per neuron

(n neurons, m timesteps) → (n fr)

(n fr) → (2 PCs)

End

PCA



Feature Extraction (Neural Coding Methods)

Temporal coding (Synchrony Index) :

Start

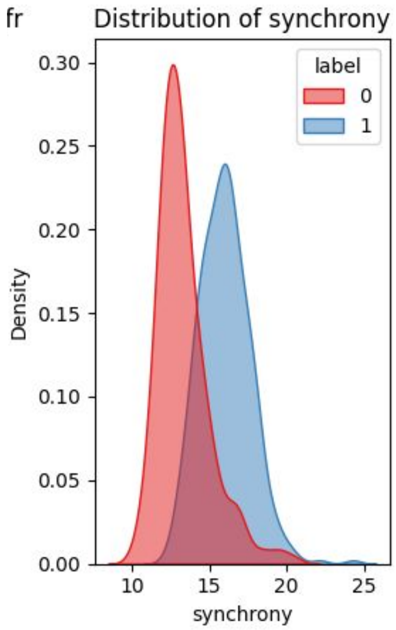
spike train per neuron → neurons pairwise synchrony

(n neurons, m timesteps) → (n , n) smoothed correlations over small time window

(n , n) pairwise synchrony → (1 avg synchrony over all neurons)

End

Here we consider only the upper-triangle of the smoothed correlation scores matrix to avoid redundancy and self-synchrony



Feature Extraction (Neural Coding Methods)

Population coding (Population Sparseness):

Start

spike train per neuron → Treves-Rolls population sparseness index

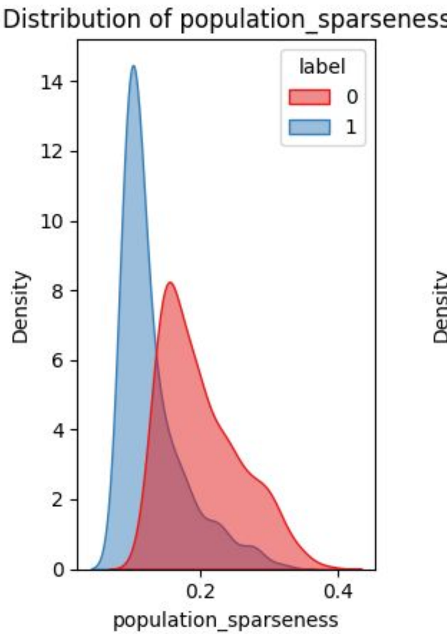
(n neurons, m timesteps) → (a)

End

Where a is the ratio of the square of the mean firing rate to the mean of the squared firing rates:

- N: number of neurons
- r_i: firing rate of neuron i

$$a = \frac{\left(\frac{1}{N}\sum_{i=1}^{N} r_i\right)^2}{\frac{1}{N}\sum_{i=1}^{N} r_i^2}$$



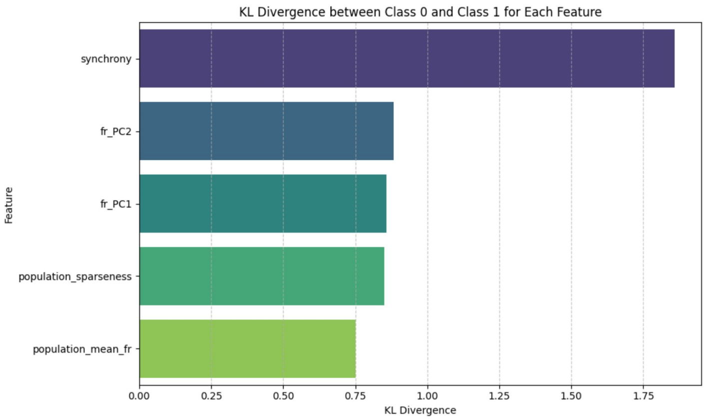
Feature Ranking and Selection - Results

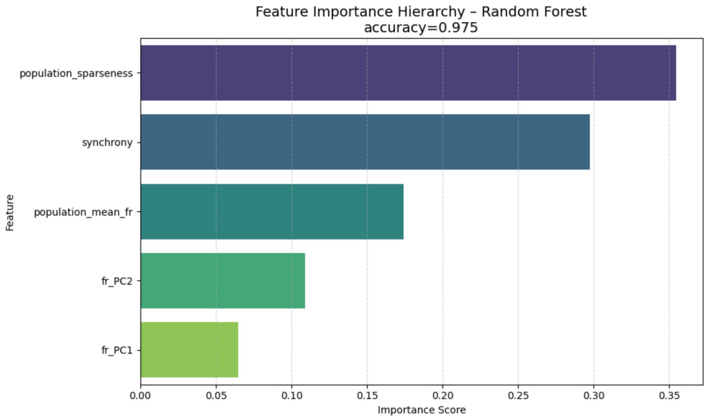
KL Divergence: Measures the difference (i.e. seperebality) of the class distributions (familiar vs unfamiliar) based on information content through its relative entropy measure.

Random Forest Importance: Ranks features based on their contribution to classification boundaries

Most informative features (both methods):
Population Sparseness and **Synchrony** scored best on both metrics., consistently showing strong discriminative power between familiar and unfamiliar stimuli.

While **Mean Population Firing Rate** and the **Fr PCs** showed to compensate for each others and generally contribute less toward classification performance.





Model Selection and Training

Models Used:

- **Logistic Regression**
  - Baseline linear model
- **Logistic Regression + Polynomial Features**
  - Captures non-linear feature interactions
- **Logistic Regression + Spline Features**
  - Flexible, smooth non-linearity
- **Support Vector Machine (RBF kernel)**
  - Strong performance on high-dimensional, non-linear data (provides a benchmark for less complex models)
- **Random Forest**
  - Robust ensemble model, provides feature importance
- **K-Nearest Neighbors**
  - Simple, non-parametric baseline

Training Setup:

- Models were trained on two distinct feature sets:
  - **All extracted features**
  - **Top 3 features** ranked by Random Forest importance

Model Evaluation Metrics

Accuracy as Evaluation Metric:

- the dataset is balanced (similar/equal number of classes, 400 each)
- we are interested in overall classification correctness.
- appropriate when both classes are equally important and misclassification costs are symmetric.

Other Evaluation Metrics:

Metrics like **F1-score** or **precision/recall** when:

- class imbalance
- false positives and false negatives have different costs, such as in medical or anomaly detection tasks.

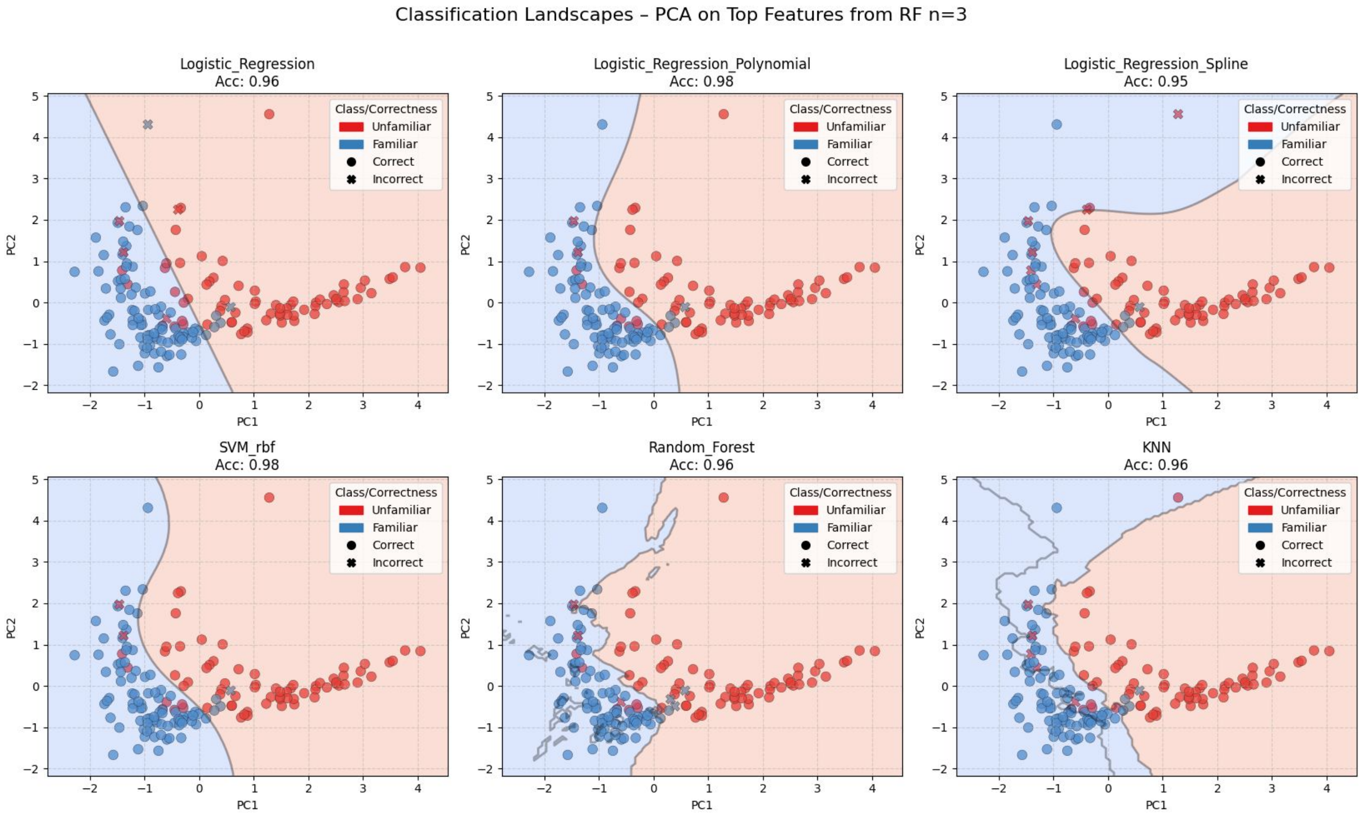
k-Fold Cross Validation:

- We Use K=5-10 Fold Cross validation
- More robust accuracy results
- Feedback about model stability through the std of the k- validation runs

Model Evaluation Metrics - Results

Selected Features:

- **Synchrony**
- **Population Sparseness**
- **Mean Population Firing Rate**

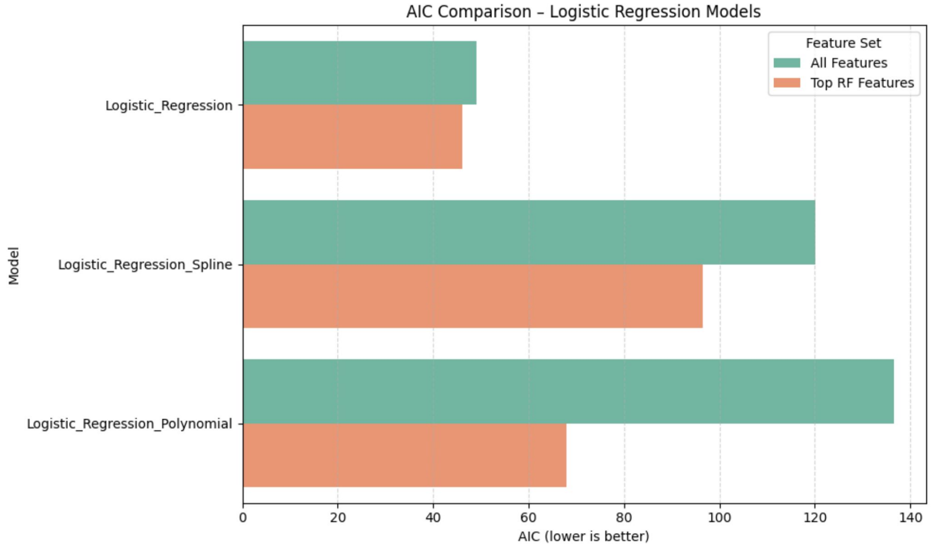


Model Comparison - Results

AIC (Logistic Regression variants only):

- Best AIC from **simple Logistic Regression** with top RF-selected features.
- **Polynomial/Spline** models, more parameters and worse AIC overfitting without proportional gain in likelihood
- Reduced feature set led to better AIC across all regressions → less redundancy.

→**Feature selection** reduces model size with minimal loss in accuracy.
→**Logistic Regression (with top features)** is most efficient (lowest AIC, high accuracy).



Model Comparison - Results

Cross-Validation (All models):

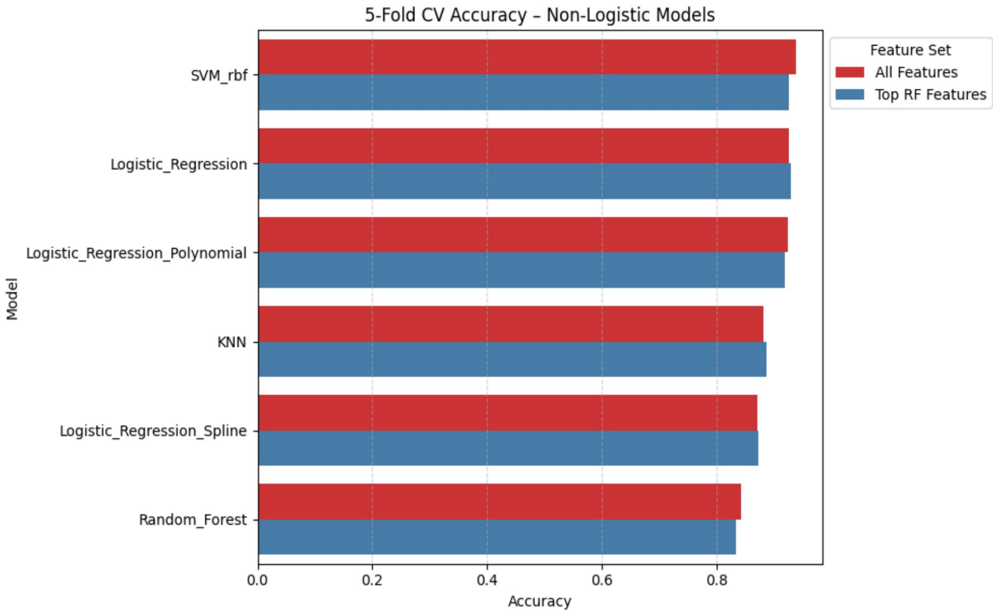
- **SVM (RBF)** achieved highest accuracy (by a small margin).
- **Logistic Regression** nearly matched SVM, especially with top features.
- **KNN and Random Forest** performed reasonably but showed more variance

→**SVM** is most accurate but more complex.

Overall:

→ small Interpretability vs accuracy trade-off: LR is preferable unless maximum performance is critical.

→ Further gains possible via optimized feature engineering or alternate encodings.



Thank You

For Additional Plots and more detailed explanations, please refer to the project-notebook:

https://github.com/esmeralda-25/NI_Classification_SpikingNeuralData/blob/main/NI_project_EsmaHodzic.ipynb